Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3301-WE01

Title: (BEng/MEng)

Structures and Geomatics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

(a) Figure Q1a shows a statically indeterminate truss structure. Assuming slender braces, determine the internal forces carried by each truss member.

[40%]

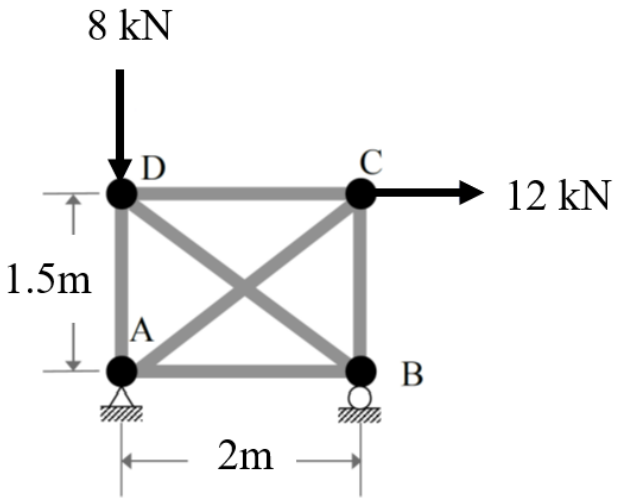


Figure Q1a

(b) Figure Q1b shows a propped cantilever beam with a roller support and carries a point load P acting at a distance x from the roller support A.

(i) Using the force/flexibility method and Table Q1, determine the value of x that gives the maximum bending moment magnitude at the fixed-end support B.

[40%]

(ii) If the point load P is replaced by a uniformly distributed load of intensity w per unit length that covers the whole span L, using the influence line method and your calculations in part (i), show that the maximum bending moment magnitude at B is $wL^2/8$.

[20%]

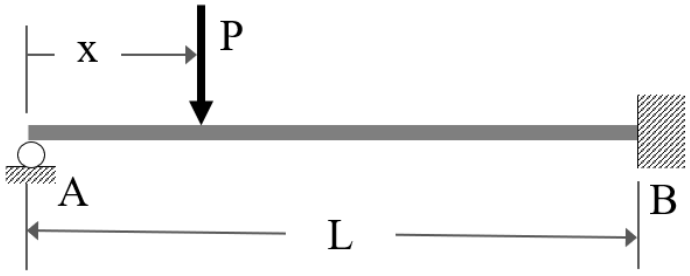
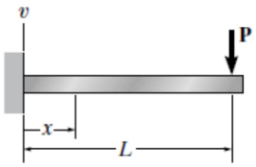
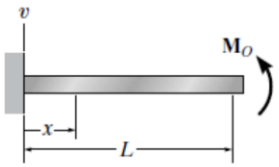
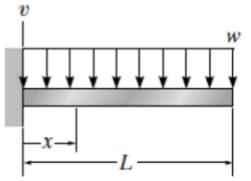
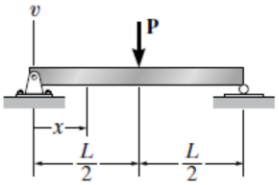
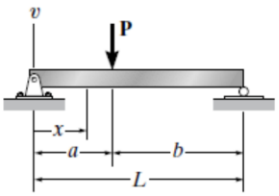
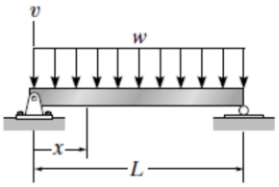
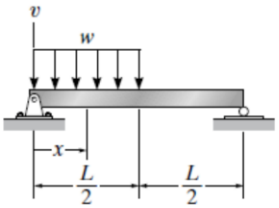
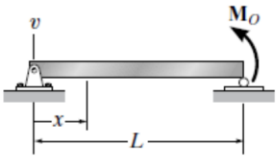


Figure Q1b

continued

Table Q1: Some standard slope/deflection solutions

Loading	$v+\uparrow$	$\theta+\curvearrowleft$	Equation $+\uparrow$ $+\curvearrowleft$
	$v_{max} = -\frac{PL^3}{3EI}$ at $x = L$	$\theta_{max} = -\frac{PL^2}{2EI}$ at $x = L$	$v = \frac{P}{6EI}(x^3 - 3Lx^2)$
	$v_{max} = -\frac{M_O L^2}{2EI}$ at $x = L$	$\theta_{max} = \frac{M_O L}{EI}$ at $x = L$	$v = \frac{M_O}{2EI}x^2$
	$v_{max} = -\frac{wL^4}{8EI}$ at $x = L$	$\theta_{max} = \frac{wL^3}{6EI}$ at $x = L$	$v = -\frac{w}{24EI}(x^4 - 4Lx^3 + 6L^2x^2)$
	$v_{max} = -\frac{PL^3}{48EI}$ at $x = L/2$	$\theta_{max} = \pm\frac{PL^2}{16EI}$ at $x = 0$ or $x = L$	$v = \frac{P}{48EI}(4x^3 - 3L^2x)$, $0 \le x \le L/2$
		$\theta_L = -\frac{Pab(L+b)}{6LEI}$ $\theta_R = \frac{Pab(L+a)}{6LEI}$	$v = -\frac{Pbx}{6LEI}(L^2 - b^2 - x^2)$ $0 \le x \le a$
	$v_{max} = -\frac{5wL^4}{384EI}$ at $x = \frac{L}{2}$	$\theta_{max} = \pm\frac{wL^3}{24EI}$	$v = -\frac{wx}{24EI}(x^3 - 2Lx^2 + L^3)$
		$\theta_L = -\frac{3wL^3}{128EI}$ $\theta_R = \frac{7wL^3}{384EI}$	$v = -\frac{wx}{384EI}(16x^3 - 24Lx^2 + 9L^3)$ $0 \le x \le L/2$ $v = -\frac{wL}{384EI}(8x^3 - 24Lx^2 + 17L^2x - L^3)$ $L/2 \le x \le L$
	$v_{max} = -\frac{M_O L^2}{9\sqrt{3}EI}$	$\theta_L = -\frac{M_O L}{6EI}$ $\theta_R = \frac{M_O L}{3EI}$	$v = -\frac{M_O x}{6EIL}(L^2 - x^2)$

continued

Question 2

(a) Figure Q2a displays a frame consisting of a single section, with uniform plastic moment capacity in all members (Mp). Calculate the ultimate force load, P, corresponding to the **composite** collapse mechanism using the dimensions shown in Figure Q2a.

[50%]

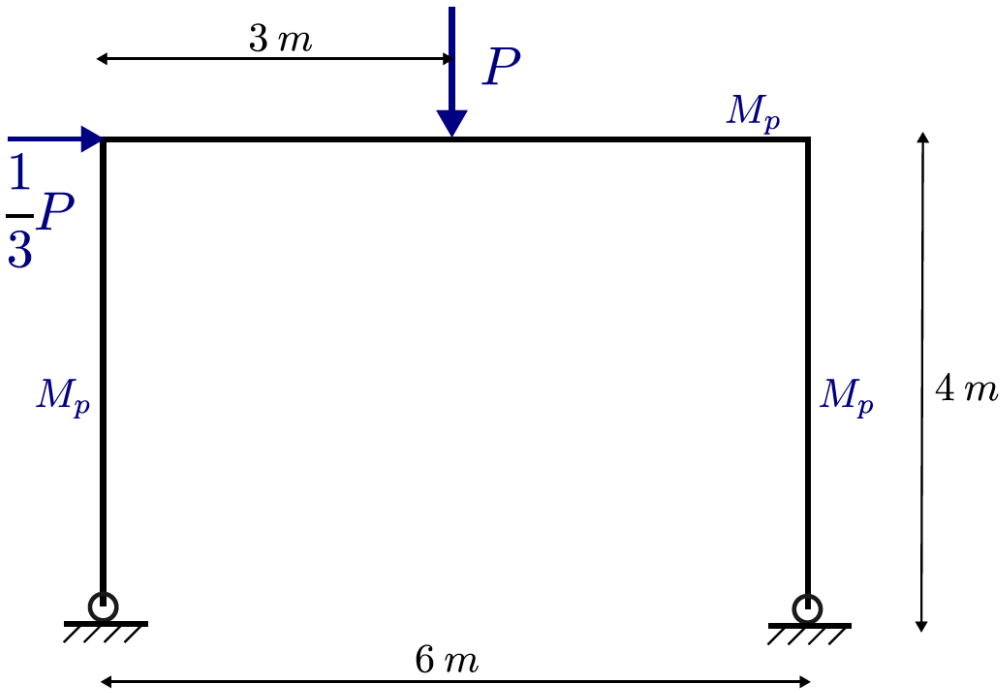


Figure Q2a

(b) Figure Q2b shows a plate fixed at two vertices, with the other two edges free. The slab is equally reinforced in x- and y-direction with the plastic moment capacity per 1m strip designated as Mp. The slab is reinforced to Mp for both hogging and sagging moments. Given the dimensions in Figure Q2b, postulate feasible collapse mechanisms and calculate the ultimate distributed pressure load corresponding to the collapse limit state of the slab. Comment briefly on your results.

[50%]

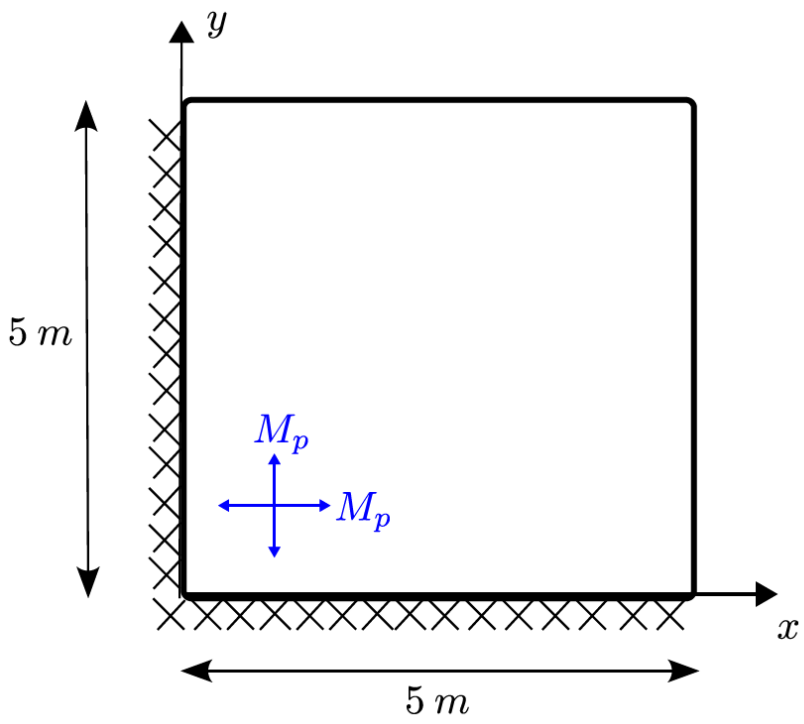


Figure Q2b

Question 3

(a) A field survey technician with an auto level equipment needs to reduce levels across a wide river. The technician is concerned about the accuracy of their measurements. Can you guide the technician on how to tackle the situation by suggesting which levelling method should be used and how good accuracy can be achieved?

[20%]

(b) Explain the concept of mass haul diagrams in the context of earthwork operations. Describe the key components and purpose of a mass haul diagram, illustrating how it aids in the planning and optimization of earthmoving projects. Provide a step-by-step explanation of how to create a mass haul diagram, including the significance of balance points and how variations in cut and fill volumes are represented.

[30%]

(c) In the field, you need to assist the site engineers in achieving the required formation level for an embankment with a slope of 2.5H:1V, as shown in Figure Q3.

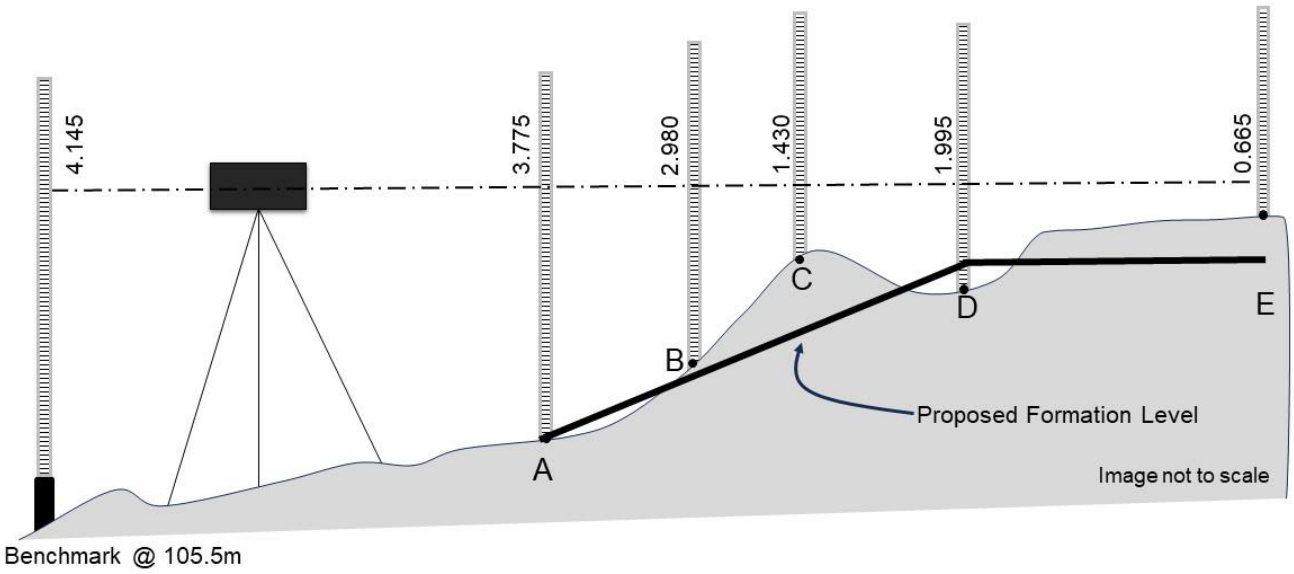
(i) Evaluate the existing ground elevations from the levelling data provided, formation levels of the embankment and the cut/fill height differences at points (A, B, C, D & E)

[40%]

(ii) Explain in detail, how do you establish elevation control in site when constructing embankment of required gradient?

[10%]

continued



Location	Distance from Benchmark (m)	Staff Reading (m)	Remarks
At Benchmark	-	4.145	Benchmark elevation is 105.5m
A	10.5	3.775	Toe of embankment. Assume the ground level is equal to the formation level.
B	12	2.980	
C	14.5	1.430	
D	20	1.995	Top of Embankment
E	25	0.665	Top of Embankment

Figure Q3

END